Booth-Assisted Mixed-Precision Reconfigurable Digital Computing-in-Memory Engine With Ternary-Input 1T2R ReRAM for Edge AI

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***Abstract*—ReRAM has emerged as a promising candidate for computing-in-memory due to its excellent compatibility with advanced CMOS processes, high density, and non-volatility. Meanwhile, Digital Computing-in-Memory (DCIM) offers higher energy efficiency than its analog counterpart and supports full-precision processing. However, existing ReRAM-based DCIMs are hindered by error-prone readout and low input parallelism. The inefficient bit-width reconfiguration also incurs significant hardware overhead during neural network mapping. This paper reports a 64Kb DCIM engine based on 1T2R ReRAM bitcells for multiply-and-Accumulate(MAC) acceleration. The ternary-input Booth-assisted multiply-in-memory (TB-MIM) flow allows bi-state ReRAM with limited readout margin to achieve high-accuracy computation with high parallelism. The corresponding reconfigurable vector adder-tree and accumulator (ReV-A^2) effectively facilitate practical hybrid-precision network mapping. Simulations show that the proposed DCIM engine achieves a normalized throughput rate of 284.44 GOPS/Kb and an energy efficiency of 730.16 TOPS/W, surpassing previous ReRAM-based DCIM designs by 3.3×.**

***Index Terms*—1T2R ReRAM, digital compute-in-memory, booth algorithm, bit-width reconfigurable.**

I. Introduction

THE explosive growth of neural network applications [1], [2], [3], [4], [5], [6] has driven an escalating demand for increased computational power and substantial memory bandwidth. Although accelerators like ASICs and FPGAs [7], [8], [9], [10], [11] have improved neural network efficiency, there remains room for further enhancements in both energy and area efficiency. A key limitation lies in the lack of robust hardware support for dynamic precision reconfiguration, which hinders the deployment of mixed-precision networks on edge devices. Computing-in-Memory (CIM) offers a promising solution by integrating data computation and storage within/near the memory array to save the energy and latency by minimizing data movement [9], [12], [13], [14]. Traditional analog CIM architectures [15], [16], face substantial challenges in supporting high-precision inference to realize large-scale extension for neural networks. By contrast, digital CIM (DCIM) architectures that leverages in-memory processing units and adder trees can effectively address the precision loss issues and provide strong resilience to process, voltage, and temperature (PVT) variations [15], [17], [18], [20]. Besides, emerging non-volatile memories like ReRAM, compared to traditional volatile memories such as SRAM, exhibit lower power consumption and higher storage density, making them ideal for delivering high-throughput, low-power network inference in resource-constrained environments [16], [17], [18], [20], [21]. ReRAM-based CIM holds promise for application in edge AI applications [18], [21],

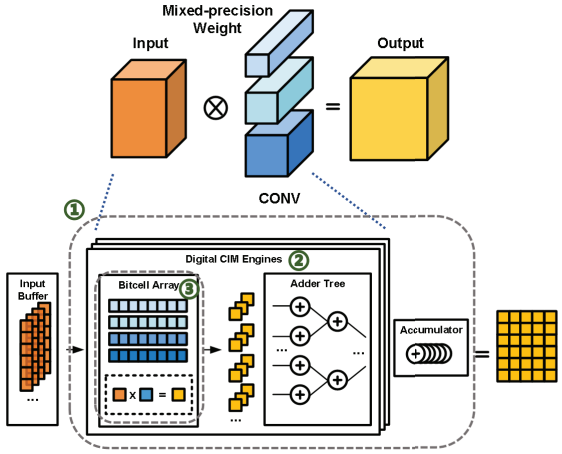


Fig. 1. Conceptual overview of dataflow in typical digital CIM engines and their key design challenges.

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[22], [23], but severe bottlenecks still remain, as illustrated in Fig. 1.

A. Inefficient Bit-Width Reconfiguration Hardware Limits Mapping Efficiency of Real-World Edge NNs

Edge devices typically utilize mixed-precision neural networks to balance computational precision and energy efficiency. This requires neural network accelerators to support efficient and dynamic bit-width reconfiguration to address varying precision demands. However, current DCIM architectures [17], [18], [20], [26], [27] largely rely on fixed bit-width adder trees, which are not optimal for accommodating different precision requirements. For instance, in 4-bit operations, 8-bit adder trees are underutilized, leading to memory inefficiencies and reduced parallelism. Conversely, 8-bit operations using 4-bit adder trees result in increased power consumption due to the need for additional full adders. Further, additional post-processing steps, such as shift-and-add circuits, can reduce overall system efficiency [17], [18]. To address these challenges, the adoption of multi-bit-width reconfigurable adder trees in DCIM architectures could offer advantages. Such designs may allow for improved parallelism in low-precision computations by fully utilizing memory cells, while potentially minimizing power consumption in high-precision tasks. This approach could enhance both the adaptability and performance of DCIM architectures, making them more applicable to real-world edge neural network scenarios.

B. Low Input Parallelism Limits Computation and Energy Efficiency of Digital CIM

Current DCIM architectures typically adopt a bit-serial input strategy [15], wherein only one bit is processed per cycle. This limits computational parallelism and energy efficiency, both of which are critical for resource-constrained edge devices. Some algorithmic optimizations, such as the Radix-4 Booth algorithm, have been proposed to reduce the number of partial products, streamline the addition process, and improve computational efficiency. In SRAM-based DCIM designs, this algorithm has demonstrated improvements by halving the number of computation cycles [28]. However, its application to non-volatile memory (NVM)-based DCIM architectures, including those based on ReRAM, remains less explored. Applying Booth multiplication in ReRAM-based DCIM architectures necessitates support for tri-state inputs, which poses challenges for conventional 1T1R arrays that require additional circuitry. In contrast, 1T2R arrays, with their symmetric structure, inherently support tri-state inputs, which may make them a suitable candidate for implementing high-efficiency in-memory Booth multiplication [29]. Integrating this algorithm into ReRAM-based DCIM designs could potentially enhance computational throughput and energy efficiency, thereby mitigating the limitations imposed by low input parallelism.

C. Small R-Ratio Limits Compute Accuracy of ReRAM Arrays

ReRAM has emerged as a promising candidate for computing-in-memory applications due to its fast write speed and scalability. However, challenges remain related to resistance variability and lifespan, which can affect performance. The 1T1R structure, while offering simplicity and high area efficiency, is subject to large read currents in low-resistance states (LRS) and high switching power, both of which can increase power consumption. Additionally, resistance drift and nonlinear I-V characteristics introduce complexities in read and write operations, especially in large arrays where voltage drops can impact multiply-accumulate accuracy, thereby increasing the need for more complex peripheral circuitry. Alternative designs, such as the 2T2R structure, mitigate some of these challenges by using differential reading, which can reduce static current and improve read margins. However, this approach comes with the trade-off of doubling the area required, which lowers memory density. The 1T2R structure offers a compromise by leveraging current summing for both readout and computation, enabling both positive and negative multiplications within a single cell [19], [30], [31]. While static current remains a challenge, a voltage-divider-based scheme utilizing 1T2R cells has been shown to reduce static current while maintaining a high readout margin with minimal area overhead [22]. This work explores the implementation of a low-power digital in-memory computing accelerator using 1T2R cells based on such a voltage-divider scheme.

To address the aforementioned challenges, this paper proposes a cross-layer co-optimization design of an compute-in-memory(CIM) accelerator based on 1T2R ReRAM. It features:

1) a low-cost, high-accuracy DCIM ReRAM array equipped with a Source Voltage Dynamic Readout Scheme (SVDRS), enabling bi-state ReRAM with a limited readout margin to perform high-accuracy computations;
2) a ternary-input Booth-assisted multiply-in-memory (TB-MIM) computation flow, utilizing a full-precision Ternary-input input ReRAM array to support low-cost radix-4 Booth algorithm, effectively halving the multiply-in-memory workload;
3) customized Reconfigurable Vector Adder-tree and Accumulator (ReV-A^2) modules, which facilitate efficient 1/4/8/16 bit-width reconfigurable partial product accumulation with minor extra hardware cost.

The remainder of the paper is structured as follows. Section II proposes the architecture of our solution. Section III details the proposed TB-MIM computation scheme, including CIM array design with 1T2R ReRAM and in-memory Booth multiplication. Section IV explains the design of the ReV-A2. Section V discusses the results and evaluations, and Section VI concludes the paper.

II. Proposed CIM Architecture

A. Proposed 1T2R DCIM ReRAM Accelerator

This section proposes a co-optimization design of a compute-in-memory(CIM) accelerator based on 1T2R ReRAM, introducing two novel methods: Ternary-input Booth-assisted multiply-in-memory (TB-MIM) and bitwidth

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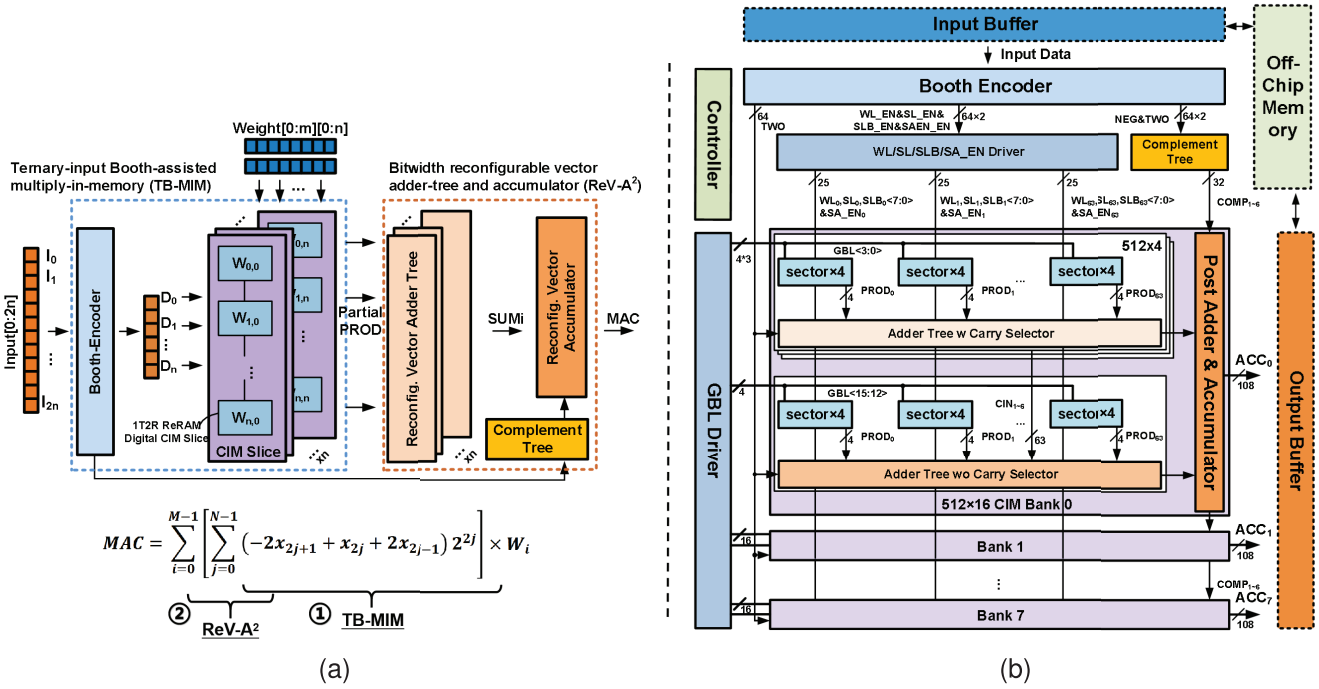


Fig. 2. Proposed 1T2R ReRAM digital CIM engine: (a) conceptual view of the event-driven MAC acceleration workflow and (b) hardware architecture.

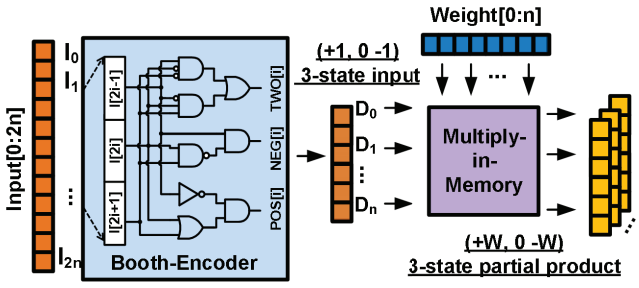
reconfigurable vector adder-tree and accumulator (ReV-A^2) (as shown in Fig. 2).

The hierarchical architecture for the event-driven MAC acceleration flow is depicted in Fig. 2a. The input data is first processed by a Booth encoder, which converts it into a format suitable for ternary operations within our customized CIM arrays. After encoding, the input is multiplied by weights pre-mapped in the 1T2R ReRAM Digital CIM array, generating partial products. By leveraging the TB-MIM scheme, ternary input multiplication is performed directly in-memory, thereby eliminating the complexity of handling negative values and optimizing computational efficiency. The partial products are then accumulated using bitwidth-reconfigurable vector adder trees and accumulators, enabling high parallelism in low-bitwidth and improved efficiency in high-bitwidth modes. Additionally, a complement tree ensures accurate handling of sign bits, resulting in correct accumulation outcomes.

As shown in Fig. 2b, the engine is composed of a controller, a Booth encoder array with 64 Booth encoders (for 128-bit inputs) and input buffers, a complement tree, a GBL driver, Bank-level WL/SL/SLB/SA_EN drivers, and eight banks. This design integrates bitcell arrays, peripheral circuits, and Booth-based dataflow optimizations, enabling high-speed, flexible, and dynamically reconfigurable bit-width acceleration for neural networks tailored to edge applications.

III. Booth-Assisted Multiply-in-Memory With Full-Precision Digital 1T2R ReRAM CIM Array

To enhance ReRAM readout accuracy and alleviate the bottleneck of low parallelism in bit-serial operations in digital CIM architectures, this section presents the design and implementation of a low-power 1T2R ReRAM array optimized for multi-layer binary weight mapping and ternary-input Booth-assisted multiply-in-memory (TB-MIM) operations. Fig. 3a illustrates the dataflow and hardware structure. This approach incorporates full-precision digital CIM arrays and a co-optimized input encoding controller to improve the efficiency of in-memory ternary multiplication, effectively doubling the MAC input parallelism.



I[2i+1:2i-1]	Pos[i],Neg[i],TWO[i]	CIM Input	CIM Output	PP[i] (IxW)
000/111	000	0	0	0
001/010	100	+1	+W	+W
011	101	+1	+W	+2W
100	011	-1	-W	-2W
101/110	010	-1	-W	-W

(b)

Fig. 3. Proposed TB-MIM's (a) dataflow and (b) configuration truth table.

A. Booth-Assisted Input Encoding Scheme

Owing to its potential for reducing the number of multiply-accumulate (MAC) operations and increasing computational parallelism, the Booth algorithm [29], [32] is widely used

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in digital multiplier design. By encoding signed input data and employing radix-4 or higher, the algorithm reduces the number of partial products (PPs), thereby improving circuit performance, power efficiency, and area (PPA). For radix-4 Booth encoding, the multiplier bits are scanned in overlapping triplets to generate control signals (Pos_i, Neg_i, TWO_i) through the Boolean expressions:

$$Pos_i = \overline{x_{2i+1}} \cdot (x_{2i} \oplus x_{2i-1}) \tag{1}$$

$$Neg_i = x_{2i+1} \cdot \overline{(x_{2i} \oplus x_{2i-1})} \tag{2}$$

$$TWO_i = x_{2i+1} \oplus x_{2i} \oplus x_{2i-1} \tag{3}$$

where $[x_{2i+1}x_{2i}x_{2i-1}]$ is the $(i+1)$-th bit triplet, with boundary condition $x_{-1} = 0$.

For MAC operations (Equation 4), the bitwise expansion of inputs and weights can be represented by Equation 5. With the introduction of the Booth algorithm, Equation 6 shows that the number of inputs is halved when using the radix-4 algorithm, leading to a reduction in the number of cycles required for in-memory multiplication. This reduction in cycles contributes to lower power consumption per operation. While the complexity of the digital circuit design may slightly increase, the overall power consumption per operation decreases.

$$\mathbf{MAC} = \sum_{i=0}^{M-1} X_i \times W_i \tag{4}$$

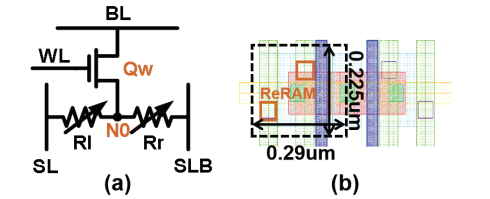
$$= \sum_{i=0}^{M-1} \left(-x_{N-1}2^{N-1} + \sum_{j=0}^{N-2} x_j 2^j \right) \times W_i \tag{5}$$

$$= \sum_{i=0}^{M-1} \left[\sum_{j=0}^{N-1} \left(-2x_{2j+1} + x_{2j} + 2\ x_{2j-1} \right) 2^{2j} \right] \times W_i \tag{6}$$

Fig. 3a presents the detailed Booth encoding circuit, while the first two columns of Fig. 3b show the input and corresponding output of the encoder. As shown in the truth table of Fig. 3b, the radix-4 Booth encoding generates partial products PP_i through the correspondence between input bit triplets $I[2i+1:2i-1]$ and the multiplicand W::

$$\mathrm{PP}_j = \begin{cases} 0 & \text{if } I[2i+1:2i-1] \in \{000, 111\} \\ +W & \text{if } I[2i+1:2i-1] \in \{001, 010\} \\ +2W & \text{if } I[2i+1:2i-1] = 011 \\ -2W & \text{if } I[2i+1:2i-1] = 100 \\ -W & \text{if } I[2i+1:2i-1] \in \{101, 110\} \end{cases} \tag{7}$$

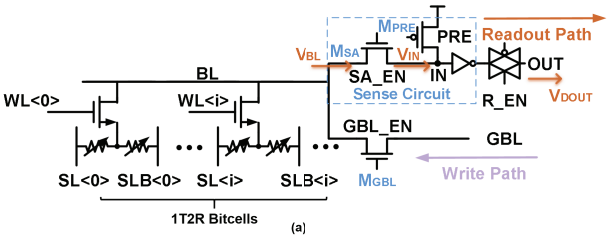
To leverage the differential programming benefits of the bitcell, the Booth encoder outputs are modified to represent POS (for +1) and NEG (for -1), instead of the conventional NEG (for sign) and ONE outputs. The POS output is activated when the partial product (PP) is W or 2W, while the NEG output is activated when the PP is -W or -2W. Notably, the NEG output in the proposed encoder behaves differently from that in traditional Booth encoders. For instance, when I[i:i-1] is 11, the NEG output in a conventional encoder is set high to simplify encoding logic, even though the partial product remains 0 due to the low ONE output. In contrast, in our design, the NEG output is set to 0 under the same conditions.



Param	Min(kΩ)	Max(kΩ)	Typ(kΩ)	3σ
R_{LRS}/R_{HRS}	20/100	5/500	10/200	30%

(c)

Fig. 4. Proposed 1T2R ReRAM bitcell's (a) schematic, (b) layout, and (c) key device parameters.



Operation	SL	SLB	SA_EN	BL_EN	WL(sel)	BL (sel)	BL (uns)
Precharge	GND	GND	GND	GND	VDD	/	/
Evaluate	Vsl	Vslb	Vsa	GND	VDD	/	/
Set Rl(Rr)	Vset/Vb1	Vb1/Vset	GND	Ven	VDD	GND	Vun1
Reset Rl(Rr)	GND/Vb2	Vb2/GND	GND	Ven	VDD	Vreset	Vun2
Forming	Vform	Vform	GND	Ven	VDD	GND	GND

*Note: Vsa = 0.45V/0V, Vsl/Vslb = 0.3V/0V, Vset =1.3V, Vreset = 1.4V, Vb1 = 0.5V, Vb2 = 0.8V, Ven =2.5V, Vun1 = 0.9V, Vun2 = 0.4V, Vform = 1.8V

(b)

Fig. 5. (a) Proposed 1T2R ReRAM CIM array structure for voltage-style in-memory program. (b) 1T2R ReRAM CIM array operation table.

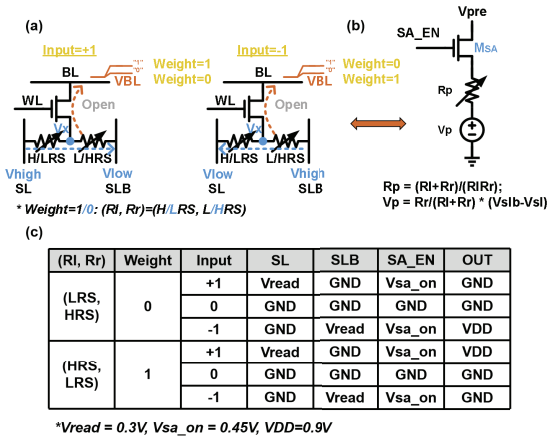
B. High Density 1T2R ReRAM for Full Precision 3-Input Multiply-in-Memory

In our proposed digital CIM array, each ReRAM bitcell stores two-state weights and three-state inputs, enabling booth-like in-memory multiplication without extra handling for negative numbers. As shown in Fig. 4a, the 1T2R ReRAM bitcell consists of a selection transistor (M0) and two symmetric ReRAM devices (Rl and Rr), with their bottom electrodes connected to a central node (N0) and top electrodes to complementary source lines (SL/SLB). This structure, storing one bit via high (HRS) and low resistance states (LRS), ensures a high readout margin and low sense current. The layout for 1T2R Bitcell for area-measurement and post-simulation is shown in Fig. 4b. As shown in Fig. 4c, this study employs multiple ReRAM resistance parameters from experimental data, with simulations accounting for up to 30% resistance drift. This ensures robust bitcell accuracy, even under adverse conditions, highlighting the structural resilience of the design.

To guarantee error-free digital compute-in-memory (CIM) operations for 1T2R cells, a low-cost multi-bit CIM array is proposed, accompanied by the low-power and full-precision Source Voltage Dynamic Readout Scheme (SVDRS).

The proposed array is of three parts (Fig. 5a): the write circuit, the storage circuit, and the readout circuit for SVDRS. For array-level interconnection, BL is configured horizontally while WL/SL/SLB are configured vertically. Multiple bitcells (e.g., 8) share a single local bitline (BL) and a readout

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(Rl, Rr)	Weight	Input	SL	SLB	SA_EN	OUT
(LRS, HRS)	0	+1	Vread	GND	Vsa_on	GND
		0	GND	GND	GND	GND
		-1	GND	Vread	Vsa_on	VDD
(HRS, LRS)	1	+1	Vread	GND	Vsa_on	VDD
		0	GND	GND	GND	GND
		-1	GND	Vread	Vsa_on	GND

*Vread = 0.3V, Vsa_on = 0.45V, VDD=0.9V

Fig. 6. Proposed 1T2R ReRAM bitcell's (a) voltage-mode readout behavior for input = +1/-1, (b) equivalent circuit during readout, and (c) relevant operation table.

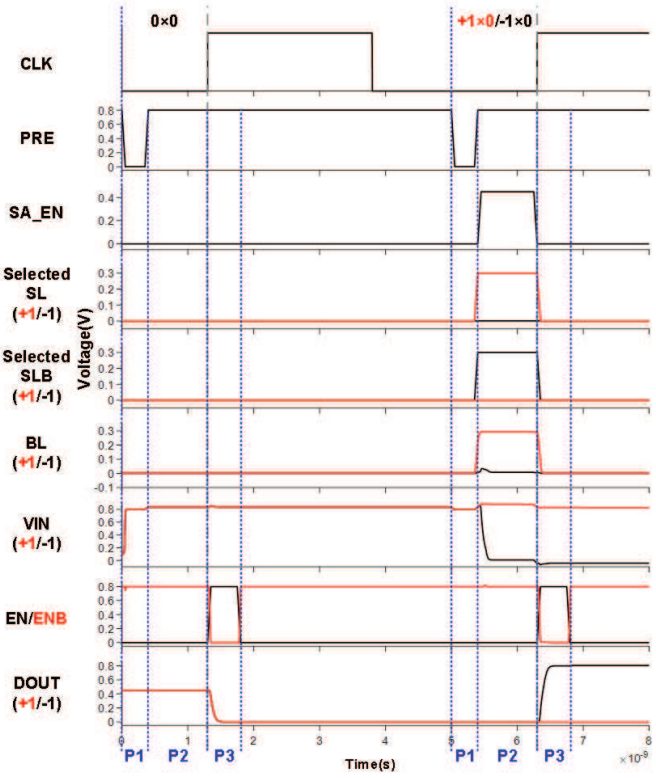


Fig. 7. The simulated waveform diagram of the CIM process.

circuit. This design enables the storage of multiple groups of weight data within a single engine, eliminating the need to reload weights when switching layers. By reducing the need for frequent write operations, this approach minimizes power consumption and latency, while also extending the endurance of the ReRAM devices. Additionally, our proposed array occupies a smaller area compared to SRAM and 2T2R ReRAM designs [22], [26], and its digital in-memory computation provides excellent scalability, allowing for large-scale, high-density weight storage at a manageable cost.

Fig. 6 further demonstrates the proposed bitcell optimization in terms of programming and readout. Fig. 6a shows the changes in BL voltage after applying voltages to SL and SLB for different stored weights, along with the corresponding sense results. To better illustrate its working mechanism, the discharge behavior of the 1T2R bitcell is represented by the equivalent circuit shown in Fig. 6b, where $R_p = (R_l+R_r)/(R_l \times R_r)$ and $V_p = R_r/[(R_l + R_r) \times (V_{slb} - V_{sl})]$. Compared to the conventional 1T1R structure, the 1T2R bitcell provides a larger readout margin, which will be elaborated on in the subsequent simulation results. Moreover, based on the differential structure and compute-in-memory (CIM) architecture, the triple-state input and bi-state weight are implemented in a single bitcell and output in the form of a complement code. The truth table and relevant operations are shown in Fig. 6c, further illustrating the significant advantages of this structure in terms of performance enhancement and operational optimization.

The simulation waveform shown in Fig. 7 illustrates two possible operations, 0×0 and $\pm\ 1 \times 0$, corresponding to the evaluation (i.e., MIM readout) process. The waveform also demonstrates the standby state pre-charge behavior between two consecutive evaluation operations within the proposed ReRAM-based compute-in-memory (CIM) array. The first subplot shows the clock signal for the digital circuits, which serves as a reference for the timing of the entire process. The readout cycles are separated by gray dashed lines, while the transitions between the three phases of the first two cycles are indicated by blue dotted lines. During simulation, R0 and R1 are in the low-resistance state (LRS) and high-resistance state (HRS), respectively, representing binary values of 0. The input for the first evaluation operation is 0, while the input for the second evaluation operation alternates between +1 (represented in red) and −1 (represented in black). The truth table for the memory and evaluation modes is provided in Fig. 5b. Each evaluation operation can be divided into three phases (P1 - P3) that correspond to key operations within the cycle. The explanation for each phase is as follows:

1) Phase 1 (P1) Is the Pre-Charge Phase: During this phase, PRE is pulled low, and V_{IN} is precharged. This prepares the circuit for the evaluation phase by ensuring that VIN reaches its initial high voltage level.

2) Phase 2 (P2) Is the Evaluation Phase: In this phase, PRE is pulled high. If the input is 0, SA_EN, SL, and SLB are grounded, keeping V_{IN} high. For inputs of −1 or +1, SA_EN is set to Vsa_on, and V_{read} is applied to SL for +1 or to SLB for −1. In the cases of $+1 \times 1$ or -1×0, the voltage on BL drops close to 0, making V_{GS} of M_{SA} higher than the threshold, which connects BL to V_{IN}. As a result, V_{IN} discharges rapidly, causing its voltage to drop. This in turn lowers the input to the inverter (INV), causing the INV output to go high. Conversely, when the voltage on BL is close to V_{read}, V_{GS} of M_{SA} remains below the threshold, so V_{IN} stays high.

3) Phase 3 (P3) Is the Output Phase: In this phase, SL, SLB, and SA_EN are all pulled low, disconnecting the storage and readout circuits. The transmission gate (TG) connects DOUT to the output of INV during this phase, which occurs at the beginning of the next clock cycle. This scheduling maximizes the allowable delay of adder trees and accumulators.

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Throughout the readout process, selected WLs are pulled up to VDD.

With 1T2R cells and the SVDRS technique, this array can inherently support tri-state inputs. The most energy-consuming cases are -1 × 0 and +1 × 1 multiplications, where V_{IN} discharges, and the output of INV is charged, requiring precharging in the next cycle. In contrast, the 0-output cases consume minimal energy, allowing input sparsity to be exploited. Even the 1 × 0 and -1 × 1 cases are power-saving, contributing to reduced power consumption. Note that the array output is in 1's complement form, with the complement +1 operation performed in the vector adder tree and accumulator.

IV. Bitwidth Reconfigurable Vector Adder-Tree and Accumulator ($ReV-A^2$)

This section further introduces the structure and functionality of the proposed ReV-A^2 scheme. In the previous section, the data front-end encoding has completed the negative in-memory multiplication, and in this section, the scheme is combined with the modified Booth multiplication to achieve high-throughput and high-resource utilization in partial product addition and multiplication accumulation. As shown in Fig. 8, the scheme includes three key components:

1) Bitwidth Reconfigurable Vector Adder Tree: Performs parallel additions with reconfigurable bit widths, dynamically optimizing the data path and enhancing efficiency. Detailed discussion is provided in later subsections.
2) Bitwidth Reconfigurable Vector Accumulator: Adjusts bit width to maximize hardware utilization during partial product accumulation, utilizing BW_MUX for dynamic scheduling. Further details are discussed in later subsections.
3) Complement Tree: Comprises 63 OR-AND gate-based compensation compressors and their compensation tree. These compressors ensure the correctness of the complement compensation while postponing the compensation operation to the later stages of the output as much as possible, reducing overall circuit power consumption and optimizing energy efficiency.

A. Reconfigurable Adder Tree Structure and Computation Flow

As shown in the Fig. 8a, similar to the vector Processing Units [8], a vector adder tree consists of several Element ATs and a top-level Head AT. The Element AT and Head AT can respectively perform fixed n-bit width additions (e.g., 4 bits) on a set of partial products (e.g., 64 partial products) (Fig. 9d). The combination of these two components can further support reconfigurable mn-bit operations (where m is an integer, and for n=4 bits, the system can reconfigure to 4/8/12/16/24/32-bits reconfiguration) (Fig. 9c). The hardware structure of the Element AT (Adder Tree with Carry Selector, ATwCS) consists of a base bit adder, an extended bit adder, a partial product shifter with shiftout, and a carry selector. The structure of the Head AT (Adder Tree without Carry Selector, ATwoCS) is similar, but to reduce hardware costs, the partial

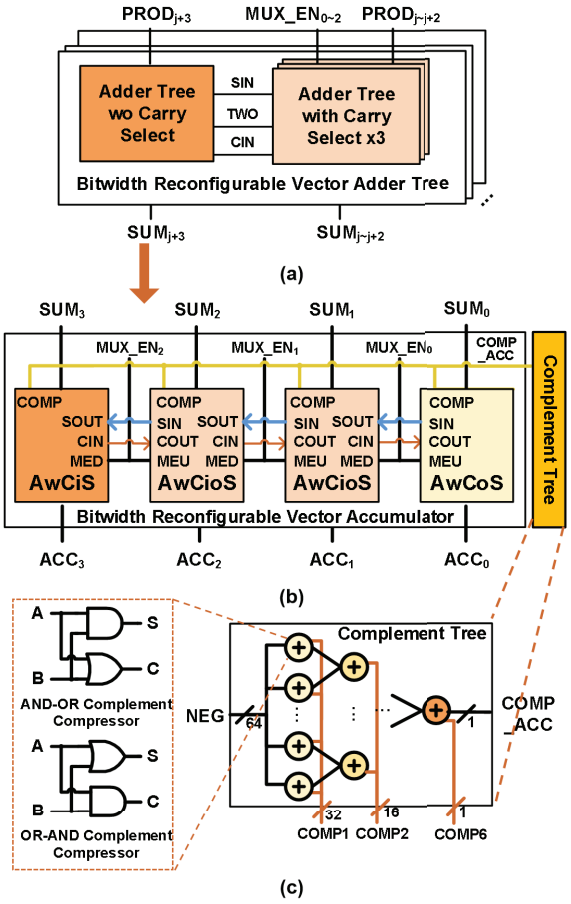


Fig. 8. Proposed 2-level hardware structure for bitwidth reconfigurable vector adder-tree and accumulator (ReV-A^2) scheme. (a) L1: 4/8/16-bit reconfigurable vector adder tree. (b) L2: Reconfigurable vector accumulator. (c) Associated full-precision Complement Tree.

product shifter does not have a shiftout, and there is no carry selector. For clarity, all adder structures referenced in this work are implemented as full adders. The detailed circuits of two partial product shifters are shown in Fig. 9a and Fig. 9b. In an ATwCS, the input stage is product shifter, which is used for TWO multiplications in radix-4 Booth implementation. If BW_MUX is low, the MSB must be extended due to sign extension. As for the lower 4 bits, they remain unchanged if TWO is 0, or they are shifted 1 bit to the left with the lowest bit of output being NEG or the carry-out of the lower ATwCS. If BW_MUX is high, the input of the extended adder tree is set to all zeros, and the MSB is shifted to the higher ATwCS or ATwoCS if TWO is high.

The Head AT handles the calculation of the highest bits $[mn-1 : mn-4]$ in the mn-bit width configuration at layer m (e.g., 16 bits, n = 4, m = 4). Since the carry does not need to propagate to higher bits, only the adder is retained without requiring the carry selector as demonstrated in Fig. 9c. The workflow of the Element AT (Adder Tree without Carry Selector, ATwoCS) is depicted in Fig. 9. Its base bit adder tree generates a carry during the addition process, which is then branched based on a selection signal.

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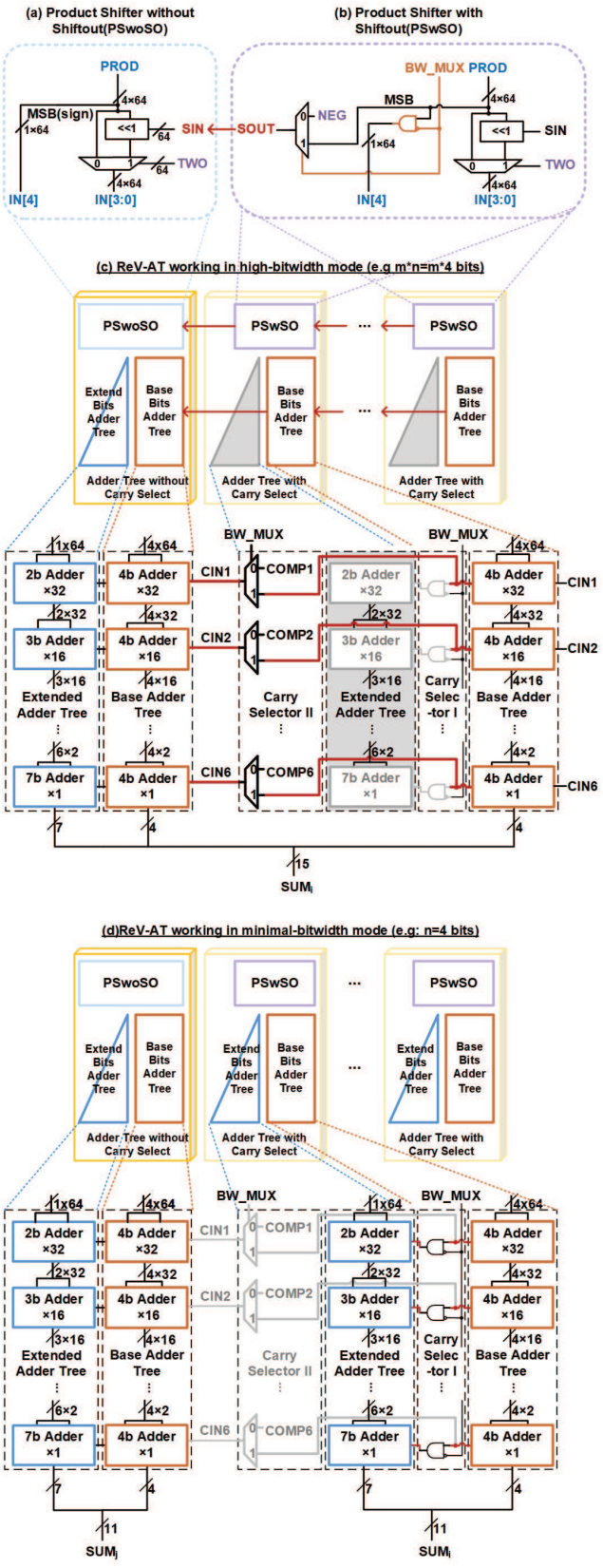


Fig. 9. Circuit diagram of the proposed ReV-AT and its operational behavior across different bit-width configurations.

During neural network deployment, the bitwidth information is compiled and stored as a bitwidth configuration signal (BW_MUX) for the AT. When the AT is enabled,

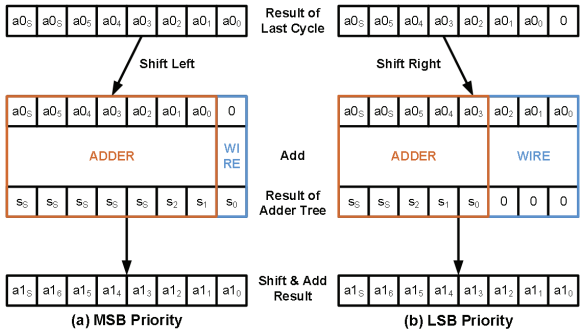


Fig. 10. Comparison of the accumulation process between the MSB priority scheme and the LSB priority scheme.

the configuration bits dynamically program the entire AT to determine the behavior of each Element AT: When the adder tree is connected with other Element ATs or the Head AT to perform higher bitwidth calculations (e.g., 8 or 16 bits), the carry is output and passed to the higher-level adder tree, and the extended bit adder tree is disabled to save power (as illustrated in Fig. 9c). Alternatively, if the adder tree does not need to output a carry to the next layer of the adder tree (e.g., when ATwoCS works independently), the extended bit adder tree is activated, and the carry from the base bits is passed along as the output complement compensation signal (COMP) (As illustrated in Fig. 9d). It simultaneously generates a 1×n bits (e.g., 4 bits) addition result.

Specifically, the carry-in of the higher ATwCS or ATwoCS is determined by the carry selector II, ensuring arithmetic correctness in the vector adder tree. The use of carry selectors incurs only a small overhead of AND gates and MUXs, which is negligible compared to the elimination of signed/unsigned input selectors and the extra adders and shifters required for cross-bank reconfiguration. Conversely, the energy savings achieved through in-bank configuration are significant, especially for high-precision operations. Additionally, this design increases the utilization of memory cells and computing parallelism for low-precision operations. It is also worth noting that the carry input of the lowest-level ATwCS is directly connected to the complement compensation signal.

B. Reconfigurable Post Vector Accumulator

Before introducing the proposed accumulator, the input order should be discussed first. Fig. 10 illustrates the accumulation process for two types of input order. $a0(a0_s \sim a0_0)$ represents the accumulation result of the last cycle. $s(s_s \sim s_0)$ represents the output of the adder tree in this cycle, and $a_1(a1_s \sim a1_0)$ represents the result of this cycle. Fig. 10a presents the process of the conventional MSB priority scheme, which is used in all published digital CIM works with timing diagrams. a_0 is shifted to the left and then added by s. The bits of s are placed from the lowest bit and its sign is extended. In this case, s_0 is directly passed to the register, while the other bits are put into the adder. In the LSB priority scheme, $a0$ is shifted to the right, and then the bits of s are needed to be placed at higher bits. Consequently, the lower bits of $a0$,

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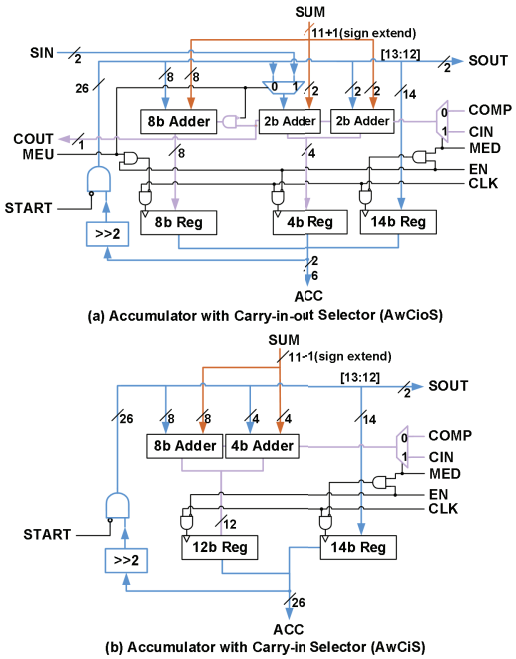


Fig. 11. Schematic of key modules for the proposed R econfigurable Vector Accumulator: (a) Accumulator with Carry-In-Out Selector (AwCioS). (b) Accumulator with Carry-In Selector (AwCiS).

which is $a0_2 \sim a0_0$ in this example, can be directly transmitted to the input of registers.

The MSB priority scheme offers advantages such as simpler accumulator design and reduced switching activities on higher bits. This is because the carry-out and shift-out directions are the same, eliminating the need to handle overflow. However, this design may require more adders in the accumulator, leading to increased latency and area overhead. In contrast, the LSB priority scheme reduces the number of adders, resulting in a shorter critical path and lower area consumption. For instance, in our design, with 4-bit or 8-bit inputs, the registers for lower bits in the LSB priority scheme do not switch, saving power. However, careful consideration is required to avoid overflow in this configuration.

Fig. 8b displays the structure of the proposed vector accumulator. The basic sub-module, accumulator with carry-in-out selector (AwCioS), is shown in Fig. 11a. (The Accumulator with Carry-In Selector (AwCiS) with its corresponding MUXs and gates removed for the highest bits is presented in Fig. 11b, since there is no carry-in for the lowest accumulator and no carry-out for the highest accumulator, respectively.) In our proposed architecture, the maximum input bit-width is 16 bits, and the outputs of the adder trees, which are also the inputs of the accumulators, are 11 bits wide. In the radix-4 Booth implementation, a MAC operation requires a maximum of eight cycles, so the input of the first cycle will be shifted by 14b in the later seven cycles. Therefore, a 14-bit register is needed for the shift-out. On the left side, the algorithm simulation has been performed and 12-bit adder and register are sufficient to prevent overflow and obtain all output results. A total of 26-bit registers and 12 full adders are required in a single accumulator. Compared with the MSB priority scheme, 12 full adders are reduced.

During the accumulation process, data streams are identified by lines with arrows indicating their directions in Fig. 11. Blue lines represent the outputs of registers, orange lines represent the outputs from the adder tree, and purple lines represent the outputs of adders inside accumulators. The enable signal of accumulators, EN, is connected to the controller. The START signal is pulled high on the first cycle of a new MAC operation to block the output of the register from being input to the adder. In this initial cycle, the adder only adds the output of the adder tree with COMP_ACC. The inputs of the adder are controlled by the START signal instead of the inputs of the registers, resulting in a shorter critical path from the arrays' outputs.

This vector accumulator can coordinate the vector adder tree to implement in-bank reconfiguration. Multiple MUXs and gates are used. In a single AwCioS, two BW_MUX inputs are required. MED is the BW_MUX from the downside (LSB side), while MEU is the BW_MUX from the upside (MSB side). Like the role of BW_MUX signals in the vector adder tree, carry-in and carry-out is controlled by these signals. In the accumulator, the shift-in and shift-out bits must also be considered. When MEU is enabled, SIN from the upside is enabled. All registers in the accumulator are clock-gated to save power. MED controls the lowest 14-bit register, while MEU controls the highest 8-bit register.

V. Evaluation Results and Analysis

The proposed architecture has been evaluated through simulations using TSMC 28nm technology. To analyze the error rate of the proposed array, Monte Carlo simulations were conducted on ReRAM devices with HRS/LRS values of 20kΩ/100kΩ, 10kΩ/200kΩ, and 5kΩ/500kΩ, with a variation of 30%, at temperatures ranging from -20°C to 100°C. At the implementation level, all digital circuits in the architecture were designed using Verilog HDL and verified through synthesis and post-simulation. To assess the hardware deployment efficiency of the bit-width reconfigurable CIM engine, we carried out experiments using the CIFAR-100 dataset and a range of neural network models, including RepVGG variants (a0, a1, a2) [24] and VGG variants (11_bn, 16_bn, 19_bn) [25]. The results from these simulations demonstrated that the proposed engine, supported by Booth-like dataflow optimizations, bitcell arrays, and peripheral circuits, offers high-speed, flexible, and dynamically reconfigurable acceleration for bit-width neural networks, making it particularly effective for edge computing applications.

A. Performance of Proposed TB-MIM Scheme

Fig. 12 demonstrates the performance of our Source Voltage Dynamic Readout Scheme. To analyze the error rate of the proposed array under extreme conditions, 1000 Monte Carlo simulations were conducted on ReRAM devices with HRS/LRS values of 20kΩ/100kΩ, 10kΩ/200kΩ, and 5kΩ/500kΩ. These simulations considered a Gaussian distribution with a 30% variation for both LRS and HRS, as well as

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LRS/HRS	Weight	VDD	Input	-20℃	-0℃	27℃	70℃	100℃
20kΩ/100 kΩ±30%	1/0	1V	-1	0.0%	0.0%	0.0%	0.0%	1.6%
			1	0.1%	0.0%	0.0%	0.0%	0.0%
		0.9V	-1	0.0%	0.0%	0.0%	0.0%	1.5%
			1	0.1%	0.0%	0.0%	0.0%	0.0%
		0.8V	-1	0.0%	0.0%	0.0%	0.0%	1.8%
			1	0.3%	0.0%	0.0%	0.0%	0.0%
10kΩ/200 kΩ±30%	1/0	1V	-1	0.0%	0.0%	0.0%	0.0%	0.0%
			1	0.0%	0.0%	0.0%	0.0%	0.0%
		0.9V	-1	0.0%	0.0%	0.0%	0.0%	0.0%
			1	0.0%	0.0%	0.0%	0.0%	0.0%
		0.8V	-1	0.0%	0.0%	0.0%	0.0%	0.0%
			1	0.0%	0.0%	0.0%	0.0%	0.0%
5kΩ/500k Ω±30%	1/0	1V	-1	0.0%	0.0%	0.0%	0.0%	0.0%
			1	0.0%	0.0%	0.0%	0.0%	0.0%
		0.9V	-1	0.0%	0.0%	0.0%	0.0%	0.0%
			1	0.0%	0.0%	0.0%	0.0%	0.0%
		0.8V	-1	0.0%	0.0%	0.0%	0.0%	0.0%
			1	0.0%	0.0%	0.0%	0.0%	0.0%

Fig. 12. Simulated error rate of ReRAM MIM scheme.

temperatures ranging from -25°C to 100°C. The customized RRAM SPICE model was based on validation data from our previous work [19] and other related studies [18], [22]. V_{DD} was varied between 0.8V and 1V. The error rates that are not 0% are marked in red. The only case with a small error rate was observed under the condition of 20kΩ/100kΩ.

Specifically, 1×1 operations at -20°C have the highest error rate of 0.3% at VDD = 1V, while -1×1 operations at 100°C have the highest error rate of 1.8% at VDD = 0.8V. These conditions are rare for edge devices. When the R-ratio is greater than 10 or the temperature is between 0°C and 70°C, the error rate is zero, ensuring the correctness of the readout circuit under typical conditions. This result is significantly better than the error rate reported in [18], demonstrating the superiority of 1T2R cells and SVDRS in terms of readout correctness, which is critical for full-precision computation in digital CIM.

B. Adder Tree and Accumulator Evaluation

All digital circuits in this architecture are implemented using Verilog HDL. They are then synthesized by Design Compiler using HVT standard cells. Clock uncertainty is set at 20% of the clock period to account for the wire delay. A Universal Verification Methodology (UVM) environment is built to verify the correctness of the digital circuit. Gate-level simulations are also performed to ensure timing correctness of the circuit and obtain accurate power switching information. Switching activity information files are generated during this process to provide accurate power consumption data for the entire engine in PrimePower. All analyses are based on a radix-4 implementation, except for the comparison between radix-2 and radix-4 implementations.

The power consumption of the proposed vector adder section with different complement trees is presented by Fig. 13. Post adder represents the compensation scheme used in [28]. The power and area overhead of the complement tree com-

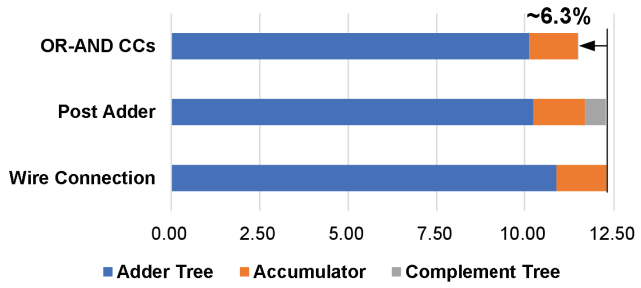
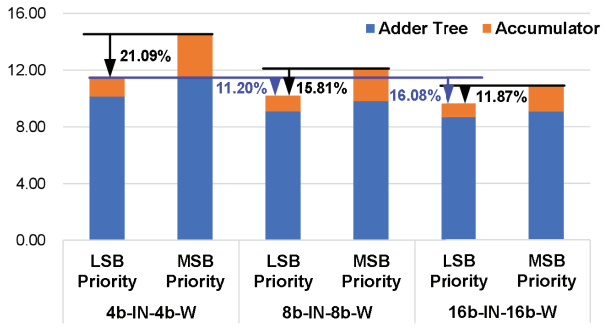


Fig. 13. The power consumption of three implementations of complement tree.



(a)

	LSB Priority(nW)	MSB Priority(nW)	LSB vs MSB(%)
Adder Tree	36409.59	37524.56	97.03
Accumulator	3517.16	5090.78	69.09
Total	33926.75	42615.34	93.69

(b)

Fig. 14. The comparative analysis of implementation for different input order in 4b/8b/16b configuration.

posed of OR-AND carry chains (CCs) is only 0.15% and 0.14% of the entire adder section, respectively. OR-AND CCs can significantly reduce the toggle rate inside the adder tree and accumulators. Compared to wire connections and the post adder scheme, the total power consumption of the adder section with OR-AND CCs is reduced by 6.3%.

Fig. 14 illustrates the analysis of both input ordering schemes across 4-bit, 8-bit, and 16-bit configurations. In terms of input order, the implementation with LSB priority consistently consumes less power compared to the MSB priority across all three configurations. This power reduction is attributed to the high frequency constraint during synthesis, where fast but power-intensive cells are required for the MSB priority implementation. In contrast, the LSB priority implementation, which uses less power-demanding cells, is more area-efficient as well. The area comparison further demonstrates that LSB priority consumes less area.

Additionally, both schemes were synthesized under a relaxed timing constraint. In the 4b-IN-4b-W configuration, the maximum delay in the MSB-first scheme was found to be 53.0% higher than that of the LSB-first scheme. This result highlights the advantages of the LSB priority scheme under high-frequency constraints. Moreover, LSB priority implemen-

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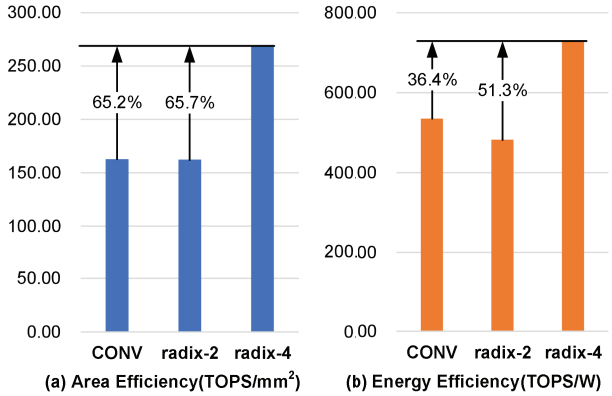


Fig. 15. The comparison of conventional bit-serial (CONV), radix-2 and radix-4 implementations.

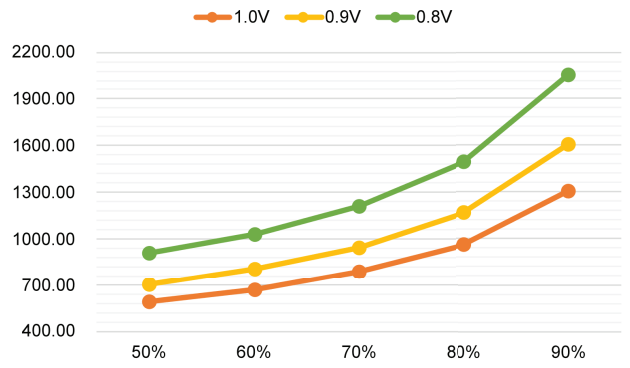


Fig. 16. The energy efficiency of the proposed design under different input sparsity and voltage.

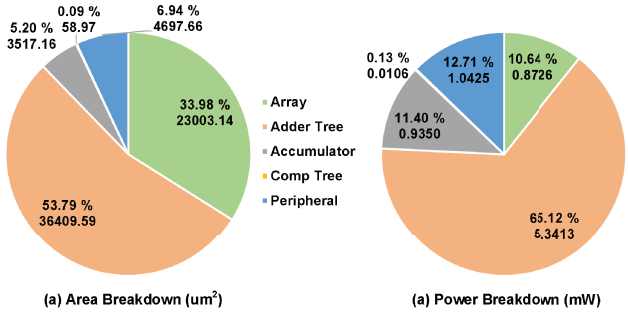


Fig. 17. The (a) area and (b) power breakdown of the proposed design at 4b-IN-4b-W, 50% input sparsity.

tations also exhibit lower power consumption in low-precision configurations.

Fig. 14 also presents a comparison of power consumption in the vector adder section across different precision levels. As previously discussed, power consumption decreases with higher precision due to the reduced number of active adders. In comparison to the 4b-IN-4b-W case, the vector adder tree and accumulator achieve energy savings of 11.20% in the 8b-IN-8b-W case and 16.08% in the 16b-IN-16b-W case. These results highlight the superiority of in-bank reconfiguration over cross-bank reconfiguration.

For the overall architecture, both radix-2 and radix-4 Booth algorithm-based engines are implemented, along with a conventional bit-serial architecture for comparison. All implementations share the same array structure, with differences only in the peripheral digital circuits. Power analysis is conducted in the 4b-IN-4b-W configuration with 50% input sparsity across all engines. Due to frequency limitations, the radix-4 architecture delivers twice the throughput of both the conventional and radix-2 architectures. Additionally, the radix-4 implementation shows a 65% higher area efficiency than the other two designs. In terms of energy efficiency, the radix-4 architecture achieves improvements of 36.4% and 51.3% over the conventional and radix-2 implementations, respectively (Fig. 15). This comparison demonstrates the significant advantages of the radix-4 architecture, particularly its ability to enhance computational efficiency by increasing input parallelism.

We also analyze the architecture's ability to exploit input sparsity. Fig. 16 presents the simulated normalized energy efficiency of the proposed engine for the 16b-IN-16b-W configuration under different levels of input sparsity and voltage. As input sparsity increases, the energy efficiency of the proposed engine improves dramatically. With 90% input sparsity at VDD = 0.8V, the engine achieves an energy efficiency of up to 2056.21 TOPS/W. This demonstrates the architecture's capability to effectively utilize input sparsity to enhance energy efficiency.

Fig. 17 shows the area and power breakdown for the 4b-IN-4b-W configuration with 50% input sparsity. The results clearly indicate that the adder tree dominates both energy and area consumption in the proposed architecture. Therefore, optimizing the adder tree is crucial for significantly improving both energy and area efficiency.

C. Hardware Deployment Performance Analysis of Neural Networks With Mixed-Precision Quantization

To comprehensively evaluate the hardware deployment efficiency of the bit-width reconfigurable CIM engine, we conducted additional experiments using the CIFAR-100 dataset and multiple neural networks, including RepVGG variants (a0, a1, a2) [24] and VGG variants (11_bn, 16_bn, 19_bn) [25]. The evaluation involves mixed-precision inference and hardware deployment validation.

The deployment framework follows a systematic workflow: First, we perform network fine-tuning and retraining in PyTorch to ensure that full-precision (FP32) inference results are on par with related works [24], [25] on CIFAR-100. Next, we define a mixed-precision quantization algorithm and conduct software-based mixed-precision inference simulations, randomly generating 2,000 bit-width configurations for layer-wise input and weight quantization. The configuration yielding the highest accuracy is selected as the optimal mixed-precision quantization (MPQ) scheme. For data reliability, we compute both Top-1 and Top-5 Accuracy. Top-1 Accuracy refers to the percentage of times the model's highest probability prediction matches the ground truth label. In contrast, Top-5 Accuracy measures the percentage of times the correct label appears

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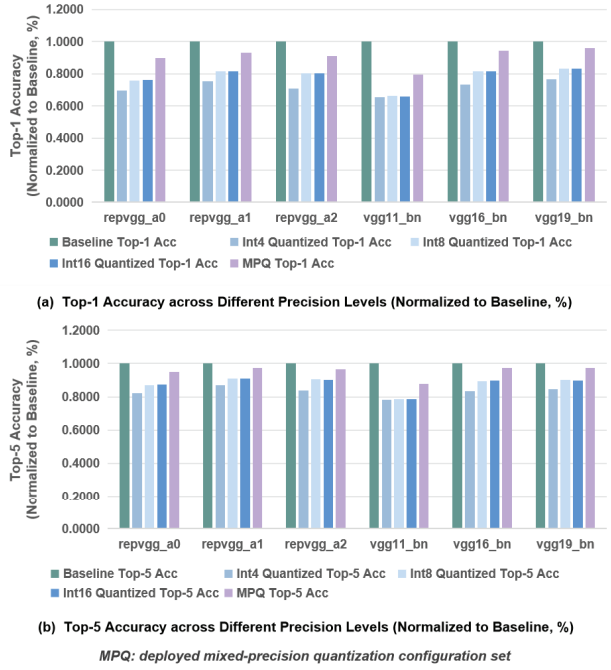


Fig. 18. Accuracy Comparison across Different Precision Levels (Normalized to Baseline, %).

within the top 5 predicted probabilities, often serving as a more relaxed metric for evaluating model performance.

Once the optimal MPQ configuration is identified, the corresponding inference results, along with those from the INT4, INT8, and INT16 baselines, are stored in bit-width-specific files. For hardware deployment, we adopt the mapping scheme proposed in [18] within the UVM platform. A verification platform reads these files and performs layer-wise operations using predefined mapping strategies. The outputs are rigorously validated against software-generated full-precision references to ensure functional equivalence. Furthermore, the measured accuracy serves as the primary evaluation metric for mixed-precision quantization. The layer-wise simulated computation latency and energy consumption are then compared and analyzed to assess the performance of the proposed configuration.

As shown in Fig. 18, mixed-precision quantization (MPQ) demonstrates significant advantages in maintaining model accuracy while reducing computational complexity. Compared to fixed-bit quantization (INT4, INT8, INT16), MPQ consistently achieves better accuracy retention across all tested networks, with Top-1 Accuracy reaching up to 95.83% of FP32 normalized accuracy (VGG19_bn). In particular, MPQ significantly improves inference accuracy in ResNet architectures, with ResNet56 recovering 57.70% of FP32 Top-1 Accuracy, far exceeding INT8 (26.89%) and INT4 (10.93%). Although VGG and RepVGG architectures exhibit greater robustness to quantization, MPQ further enhances Top-1 Accuracy while reducing bit-width requirements, achieving precision closer to INT16 while maintaining lower computational costs than INT8. Notably, for large-scale models such as RepVGG_a2 (26.94M parameters, 1850.22M MACs), MPQ retains 91.24% of FP32 Top-1 Accuracy while optimizing bit-width efficiency.

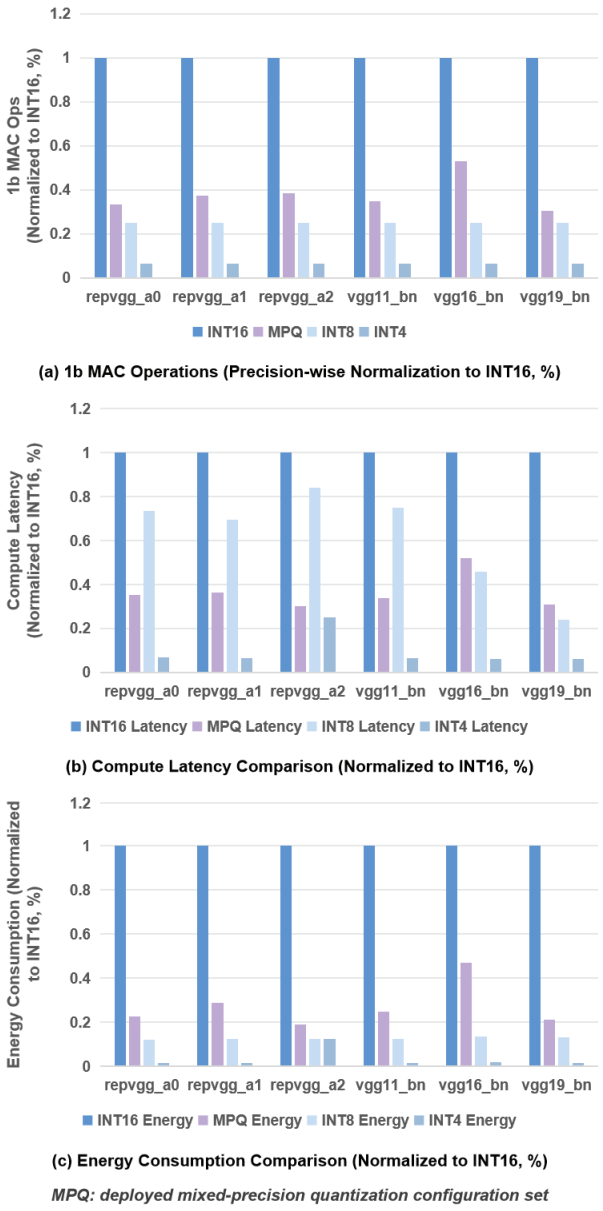


Fig. 19. Hardware Mapping Perfromance (Normalized to INT16, %).

In summary, MPQ achieves an optimal balance between inference accuracy and computational efficiency, making it an ideal solution for hardware deployment.

As shown in Figure 19, the mixed-precision inference implemented on the proposed bit-width reconfigurable DCIM engine demonstrates superior computational efficiency, inference latency optimization, and energy reduction compared to fixed-bit quantization (INT4, INT8, INT16), achieving a well-balanced trade-off between performance and accuracy.

In terms of computational cost (Figure. 19a), the MPQ scheme deployed on this engine reduces 1-bit MAC operations to approximately 34.93% of INT16, with a maximum reduction of 69.48% in VGG19_BN. Notably, in large-scale models such as RepVGG_a2, MPQ achieves only 38.54% of INT16's computational cost, significantly lowering computational overhead while maintaining high accuracy. Compared to INT8

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TABLE I
COMPARISON WITH STATE-OF-THE-ART COMPUTING-IN-MEMORY WORKS

	TCAS I 2022 [22]	TCAS II 2023 [31]	TCAS I 2022 [17]	TCAS II 2023 [18]	ISCAS 2023 [20]	This Work
Technology	40nm	28nm	65nm	28nm	28nm	**28nm**
Cell Structure	1T2R ReRAM	1T2R ReRAM	1T1R ReRAM	1T1R ReRAM	1T1M MRAM	**1T2R ReRAM**
Cell Area (um^2)	0.1564	0.06825	0.847	0.3057	0.25	**0.06525**
Capacity	32Kb	384Kb	64Kb	16Kb	64Kb	**64Kb**
Weight Precision	9b	8b	8b	1b	8b	**4b/8b/16b**
Operation Mode	Analog	Analog	Digital	Digital	Digital	**Digital**
Implementation	Pre-layout	Pre-layout	Pre-layout	Post-layout	Pre-layout	**Pre-layout**
Normalized Throughput[*1] (GOPS/Kb)	6.4	-	85.33	51.2	102.4	**284.44**
Energy Efficiency[*1] (TOPS/W)	1625	400.2	218.24	1966[*2]	203.44	**730.16**[*3] $\sim$ **2056.21**[*4]
Area Efficiency[*1] ($TOPS/mm^2$)	6.4	-	-	74.47	-	**161.37**[*5]

*Note: 1. Normalized to 1b operations.
2. Performance in 90% input sparsity, 0.6V.
3. Performance in 4b-IN-4b-W case, 50% input sparsity, 0.8V.
4. Performance in 16b-IN-16b-W case, 90% input sparsity, 0.8V.
5. Estimated under 60% layout utilization.

quantization, which is fixed at 25% of INT16 computations, MPQ achieves similar computational savings while offering significantly higher accuracy.

Regarding inference latency (Figure. 19b), the MPQ deployment effectively reduces computation time to 30.98%-52.17% of INT16, significantly accelerating inference. In particular, RepVGG_a2 achieves a 69.85% reduction in latency, highlighting MPQ's efficiency in optimizing inference speed. It is noteworthy that MPQ outperforms INT8 quantization in most networks. For instance, in VGG16_BN, MPQ achieves 52.17% of INT16's latency, whereas INT8 remains at 45.65%, demonstrating superior latency control. Although INT4 achieves the lowest latency, its excessive accuracy degradation limits its practical applicability. In contrast, MPQ maintains high accuracy while reducing latency, further reinforcing the optimization capabilities of the proposed engine.

In terms of energy consumption (Figure. 19c), the MPQ scheme significantly reduces inference power consumption to 19.07%-46.79% of INT16, with a maximum reduction of 80.93% in RepVGG_a2 and 78.71% in VGG19_BN, greatly improving energy efficiency. Compared to INT8, which typically consumes 12%-13% of INT16's energy, MPQ retains significantly higher inference accuracy while maintaining a similar power footprint. While INT4 further reduces power consumption, its substantial accuracy loss makes it impractical for deployment, reaffirming MPQ as the optimal choice for balancing power efficiency and accuracy.

The test results demonstrate that the mixed-precision neural network can fully leverage the in-memory computing acceleration circuit proposed in this work to enhance computational efficiency and energy performance. By allowing a slight reduction in inference accuracy, mixed-precision networks significantly reduce computational complexity and parameter storage requirements. Compared to low-precision networks, mixed-precision networks achieve substantial accuracy improvements with minimal computational and storage overhead. These findings highlight the critical role of bit-width reconfigurability in digital in-memory computing neural network accelerators.

Table I compares our proposed architecture with state-of-the-art CIM works [17], [18], [20], [22], [31]. It is important to note that all throughput-related data is normalized to 1b operations to ensure a fair comparison. Utilizing the radix-4 Booth implementation and an advanced technology node, both the normalized throughput and energy efficiency are improved by 3.3× compared to previous work with similar multi-bit architectures [17]. Additionally, the normalized throughput and energy efficiency are 2.7× and 3.6× higher than those of the MRAM-based engine [20].

For high input sparsity, the energy efficiency of the proposed engine slightly exceeds(1.2×) that of the engine with approximate computation [18], and the area efficiency is 2.3× higher due to the radix-4 implementation and higher operating frequency. When compared to SRAM-based digital CIM designs [34], the proposed engine maintains competitive normalized throughput and area efficiency, despite the inclusion of multiple cells per read circuit. Although the ReRAM-based engine exhibits relatively low energy consumption, it can be turned off during standby periods to further reduce power consumption.

VI. CONCLUSION

This work introduces a 28nm 1T2R ReRAM-based digital CIM engine optimized for edge computing. The TB-MIM scheme ensures reliable high-accuracy operation under various conditions, while the engine's reconfigurability and high hardware utilization, enabled by the ReV-A^2 structure, make it highly suitable for the deployment of flexible, modern NNs. The engine achieves significant improvements in both normalized throughput and energy efficiency, with at least 3.3× and 3.6× increase, respectively. These advancements highlight the engine's potential as an efficient solution for resource-constrained, on-device AI processing.

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